Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack Software Engineer with 2+ years shipping **React**, **TypeScript**, and **Python** products, backend APIs, and **data pipelines**, applying LLMs to production workflows and partnering with non-technical stakeholders.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data integration pipelines** syncing FastAPI services, PostgreSQL, SQL Server, and Supabase with incremental loads, reconciliation checks, and CI/CD-backed releases, processing 5K+ monthly jobs at 99%+ reliability.
- Designed a multi-tenant platform with normalized schemas and **row-level security** for 500+ users, cutting dashboard latency from 1.2s to 350ms, and restructured async **LLM orchestration** processing 1K+ jobs at 3x throughput.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack grading platform with a **React** dashboard and Flask REST APIs for upload, rubric generation, and grading history, serving 150+ instructors and 10K+ daily requests on AWS.
- Integrated GPT-4o, Claude, and Qwen-VL behind a switchable **LLM provider layer** with OCR fallbacks and per-call cost tracking, cutting instructor feedback turnaround 60% across concurrent batch grading workloads.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Redesigned **ETL pipelines** on AWS Glue and S3 that consolidated user activity logs from multiple sources into normalized formats for analytics and APIs, improving pipeline efficiency 68% and cutting storage costs 15%.
- Partnered with Microsoft and GitHub stakeholders in Agile sprints from **requirements through deployment**, debugging API integrations and adding pagination and caching to improve response times 30% for low-bandwidth users.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **React and TypeScript** resource-management platform on Node.js microservices for authentication, role-based access, customer records, billing, and lead management, syncing 2M+ records across locations in real time.
- Modeled Azure SQL data for users, resources, invoices, and leads, and used **analytics** on feature usage and drop-off points to guide fixes that cut resource wastage 42% and lifted lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed the end-to-end architecture, mapped 12+ integration points, and **presented it to the client**, then built a React and TypeScript web dashboard with Flask booking APIs under 200ms.
- Built hourly **Spark** micro-batch pipelines on Databricks over 50K+ booking and payment events, aggregating peak hours, active users, and revenue summaries for internal dashboards and client reporting.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS
Frontend: React, Next.js, React Native, TailwindCSS, WebSockets
Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, Microservices, Role-Based Access, JWT
Data: PostgreSQL, SQL Server, Azure SQL, MongoDB, Supabase, Data Modeling, ETL Pipelines, Spark, Databricks, AWS Glue
AI, Cloud & Practices: LLM Systems, Prompt Engineering, AWS, Azure, Docker, CI/CD, GitHub Actions, Git, Agile, Requirements Analysis

Projects

Organization Graph Platform | *TypeScript, React, Node.js, PostgreSQL, Python, Docker*

- Designed a **TypeScript** platform that syncs HR, recruiting, and issue-tracker data into a unified people, team, and project graph with incremental ETL, entity resolution, field-level permissions, and a React **org chart** explorer.

Agent-Ready Company Data Service | *Python, FastAPI, MCP, PostgreSQL, Embeddings*

- Designed a FastAPI and **MCP** service exposing people, team, and work-item data to AI agents through typed tools with per-user authorization filters, audit logging, and an evaluation set for **agent answer accuracy**.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Twilio, TailwindCSS*

- Built a **TypeScript** hospital operations platform with doctor and hospital dashboards, appointment workflows, consent-based record access, and Twilio alerts, winning Smart India Hackathon and reducing administrative workload 30%.